dark matter

standard model

THEORY & FUTURE OF DARK MATTER SEARCHES

dark matter

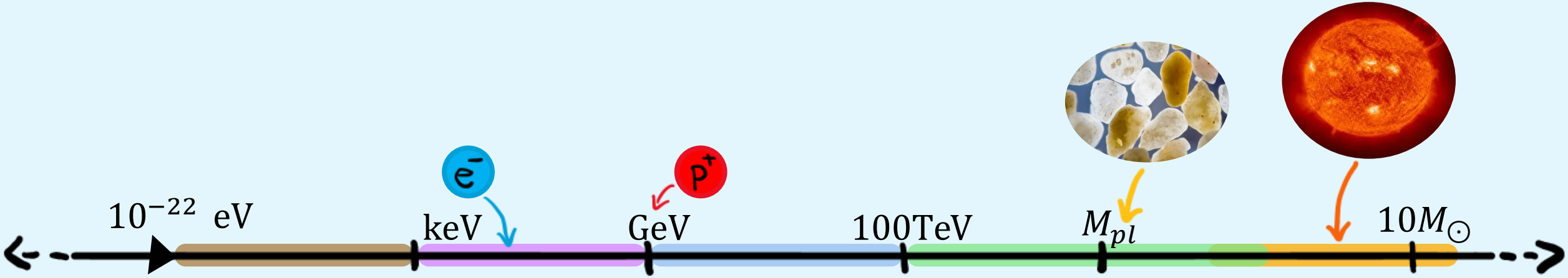
standard model

YILDA BOUKHTOUCHEN

DARK MATTER DAY MASTERCLASS 2024

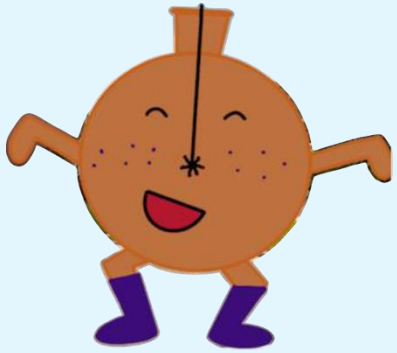
WHAT IS DARK MATTER?

WHAT ~~IS~~ COULD DARK MATTER BE?

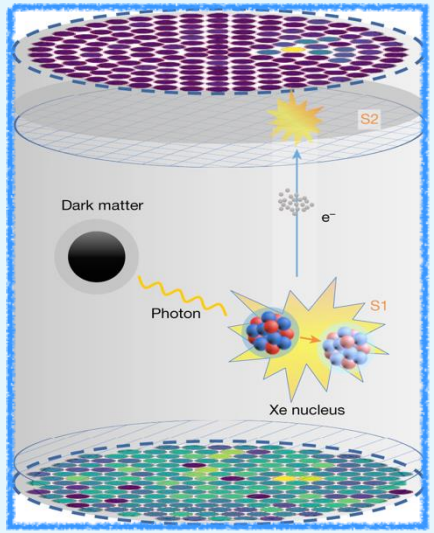


"ultralight" dark matter

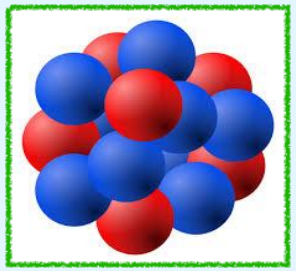
"light" dark matter



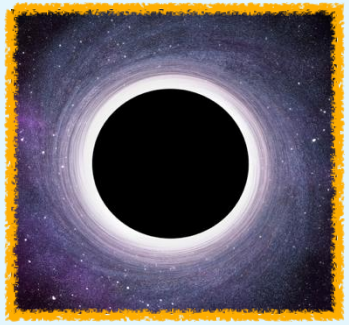
WIMPs



composite dark matter



primordial black holes



Scale Credit: Tongyan Lin

WHAT DO WE KNOW SO FAR?

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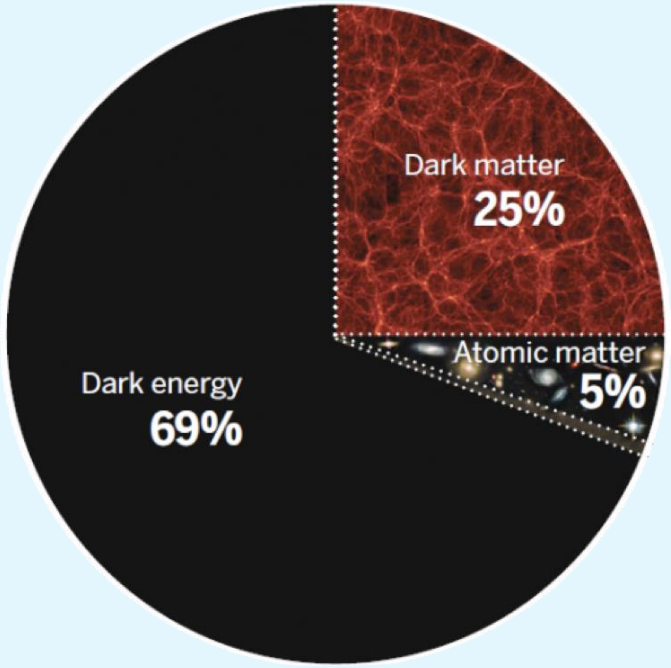

Science/AAAS

abundance

5 times as much
as “normal stuff”

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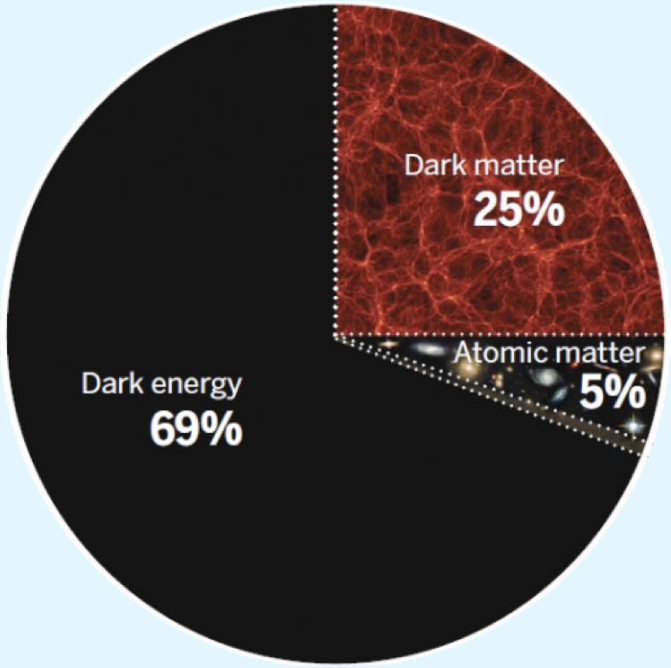

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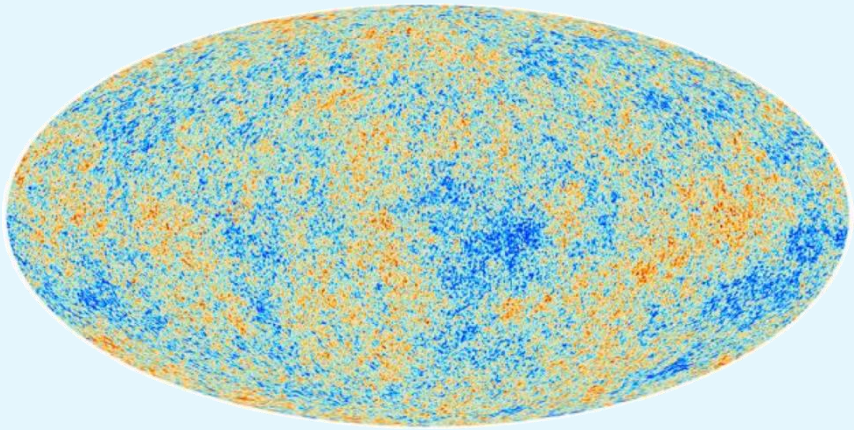

ESA & Planck

abundance

5 times as much as “normal stuff”

stable particle

has been around since early universe, and is still around.

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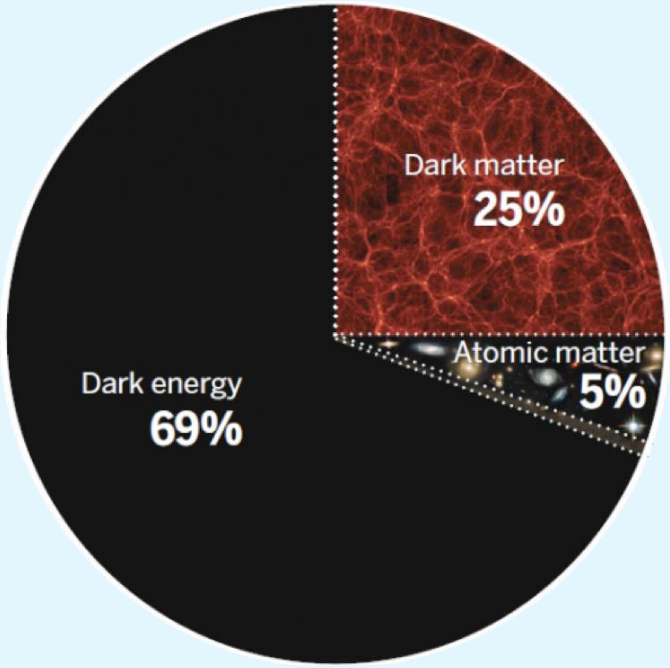

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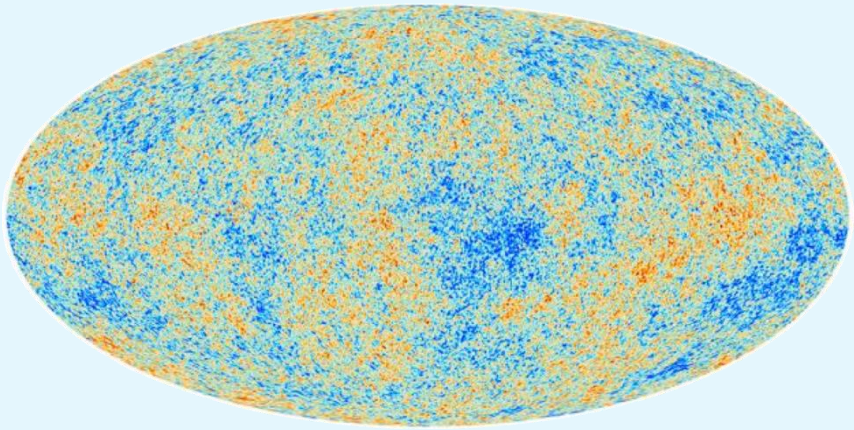

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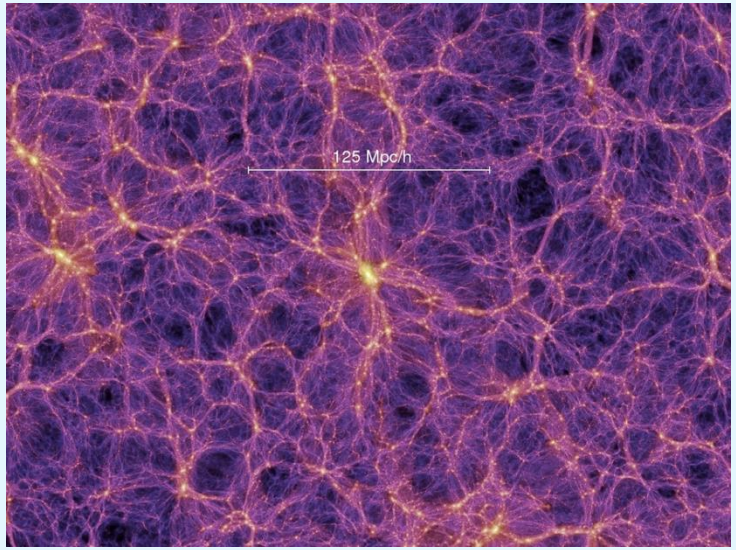


Millenium Simulation

cold, collisionless

does not interact with itself or with photons

SO WHAT IS THE STRATEGY?

a greater physical hypothesis or motivation

what is our dark matter model?

interactions with SM particles?
how often?
how was it created?

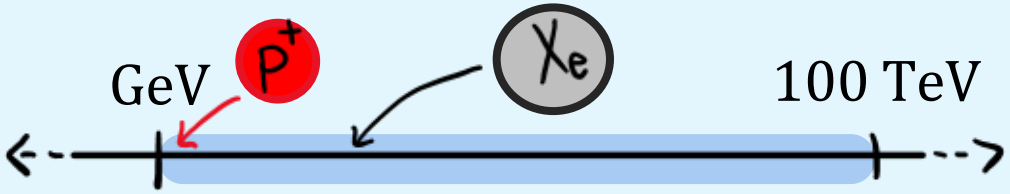
does it fit with observations?

- ☑ stable?
- ☑ the right amount?
- ☑ we didn't break the universe?

how to best look for it?

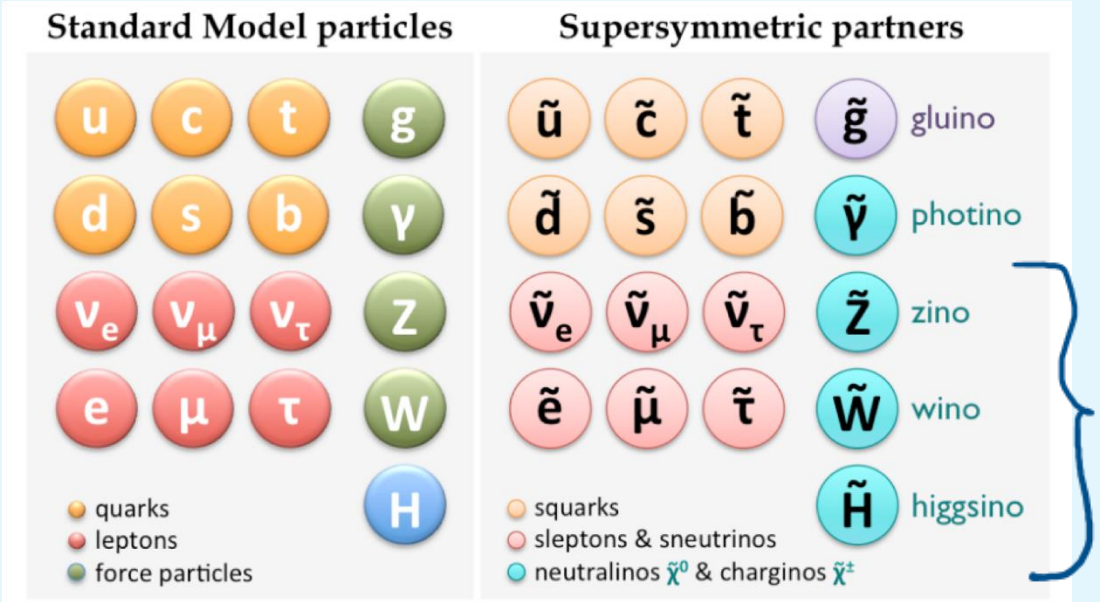
which target particles to use?
what does an interaction look like?
what kind of detector?

EXAMPLE: THE WIMP "MIRACLE"



WEAKLY **I**NTERACTING **M**ASSIVE **P**ARTICLE

theoretical motivation: SUPERSYMMETRY



every particle has a supersymmetric partner

neutralinos are the partners of gauge bosons.
they would interact with the **weak** force

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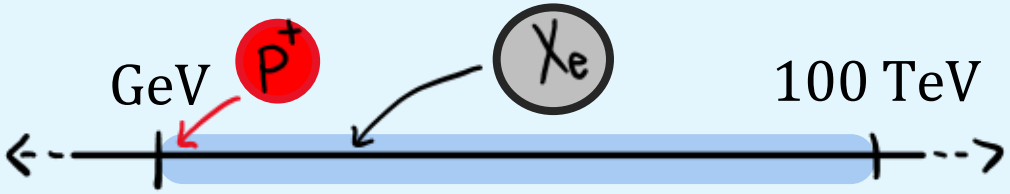

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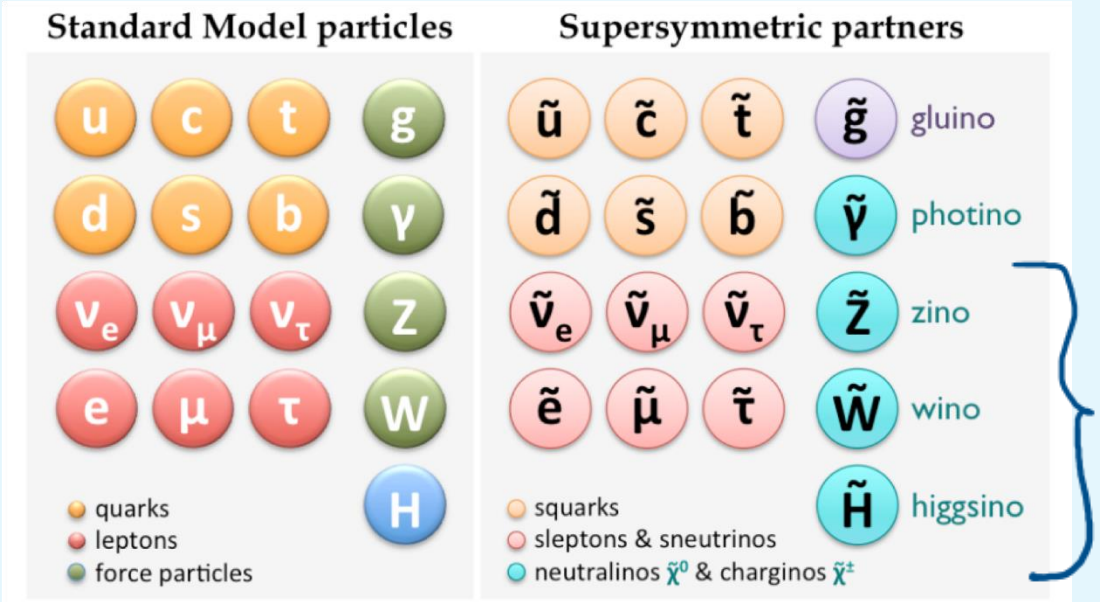


every particle has a supersymmetric partner

neutralinos are the partners of gauge bosons.
they would interact with the **weak** force

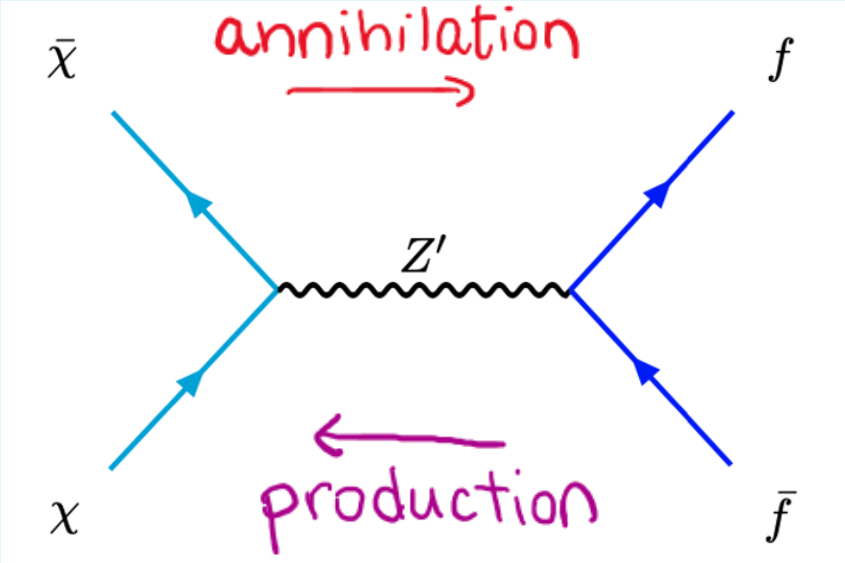
- ☑ stable
- ☑ but do interact
- ☑ interact very rarely → "cold"

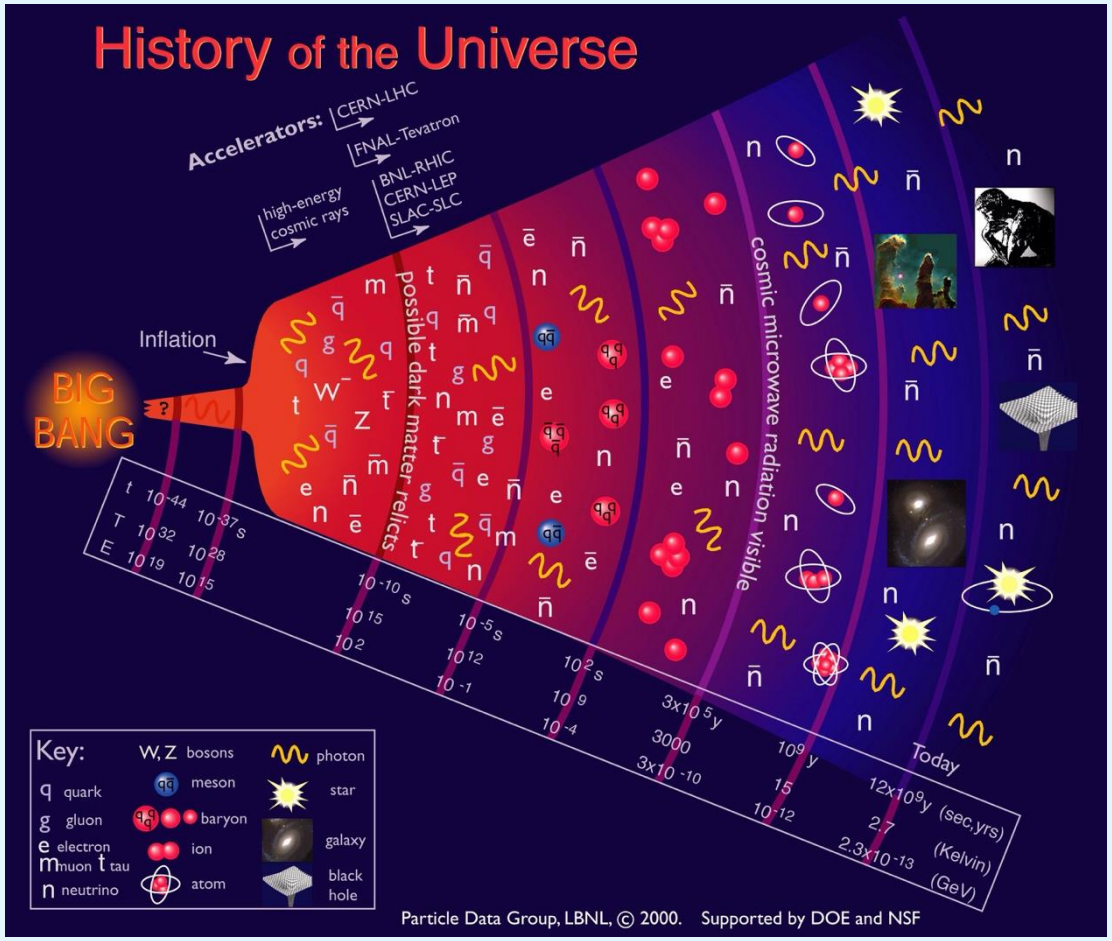
it would be a great WIMP!

DOES IT FIT OBSERVATIONS?
HOW WAS IT PRODUCED?

the universe began **very, very hot**.

high temperature = **high energy particles do all sorts of interactions!**





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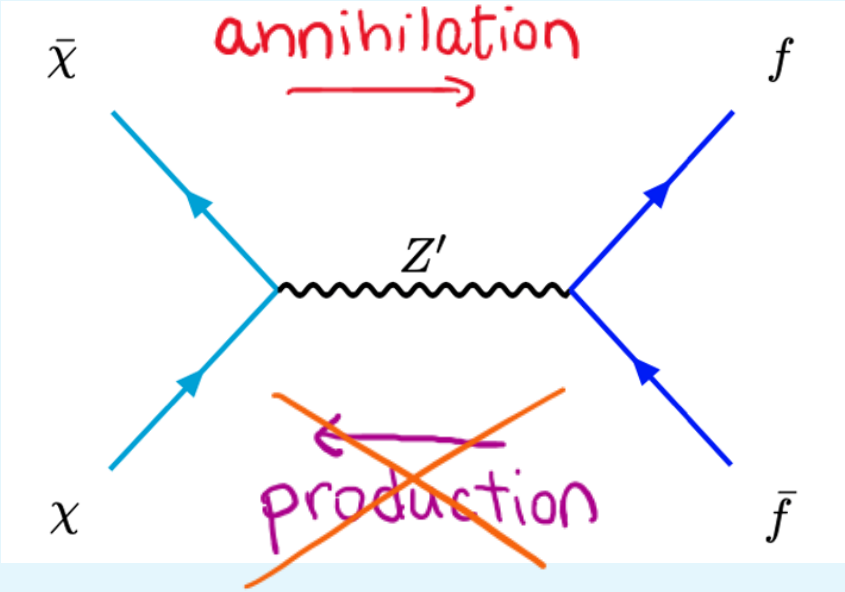
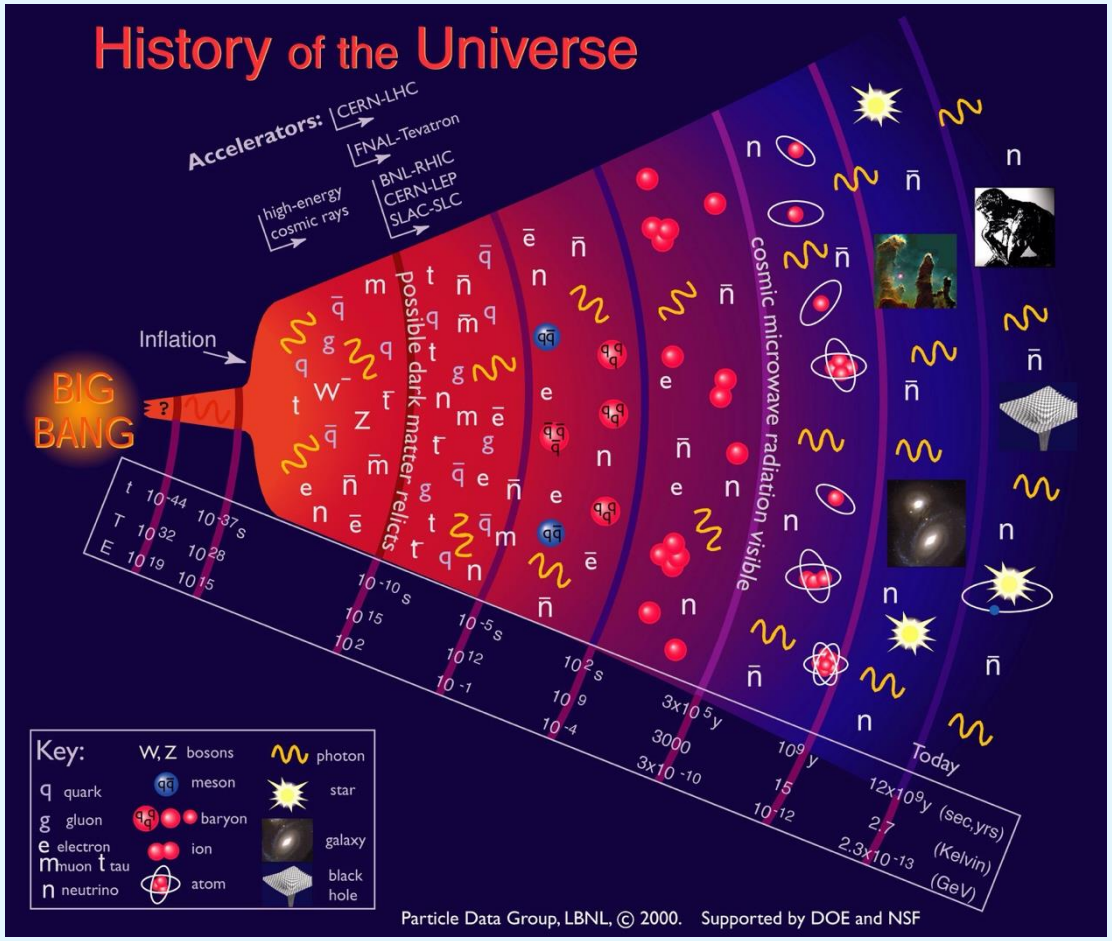


$$E = mc^2$$



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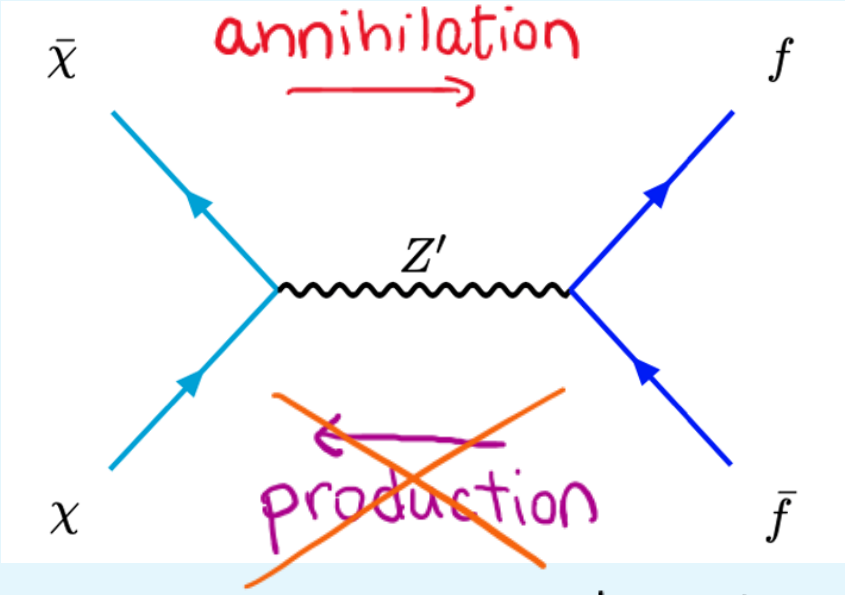
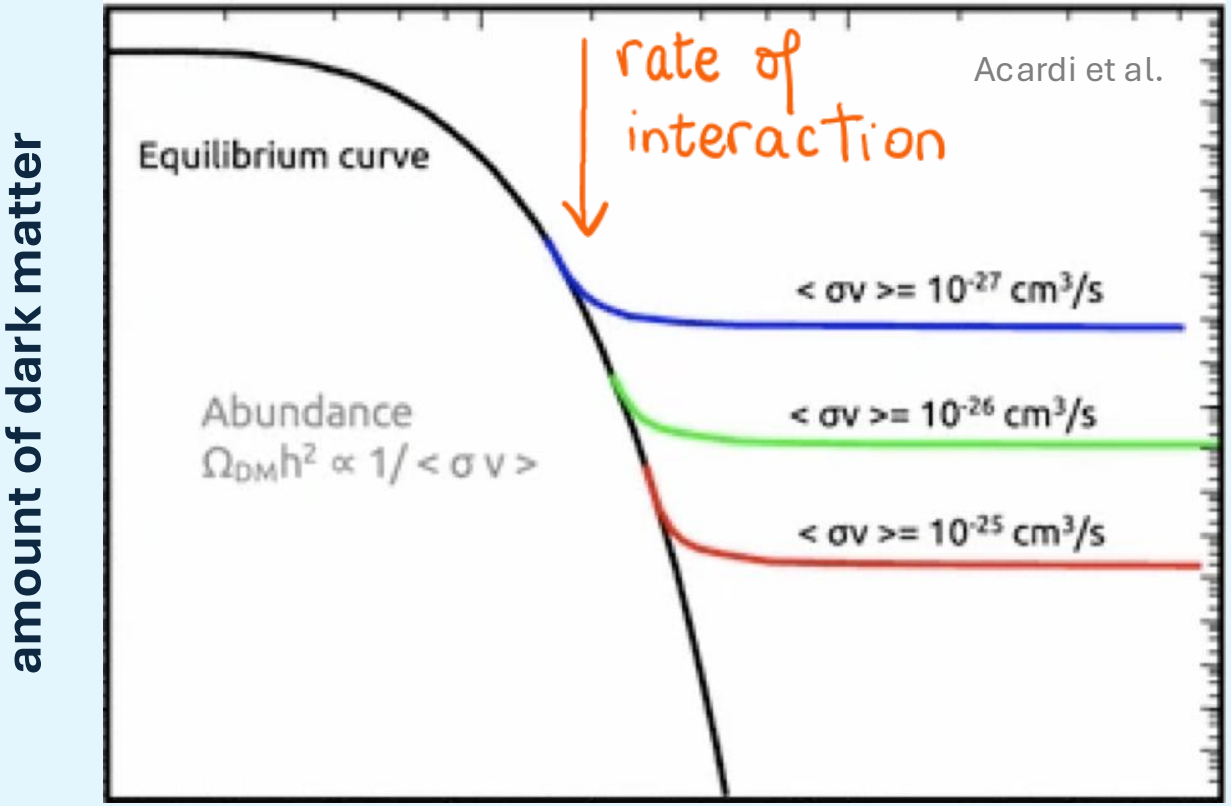


$E = mc^2$ depends on mass



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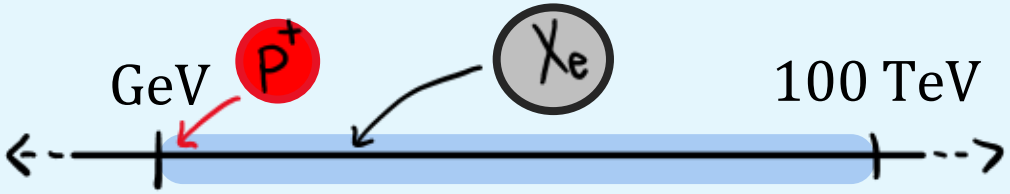

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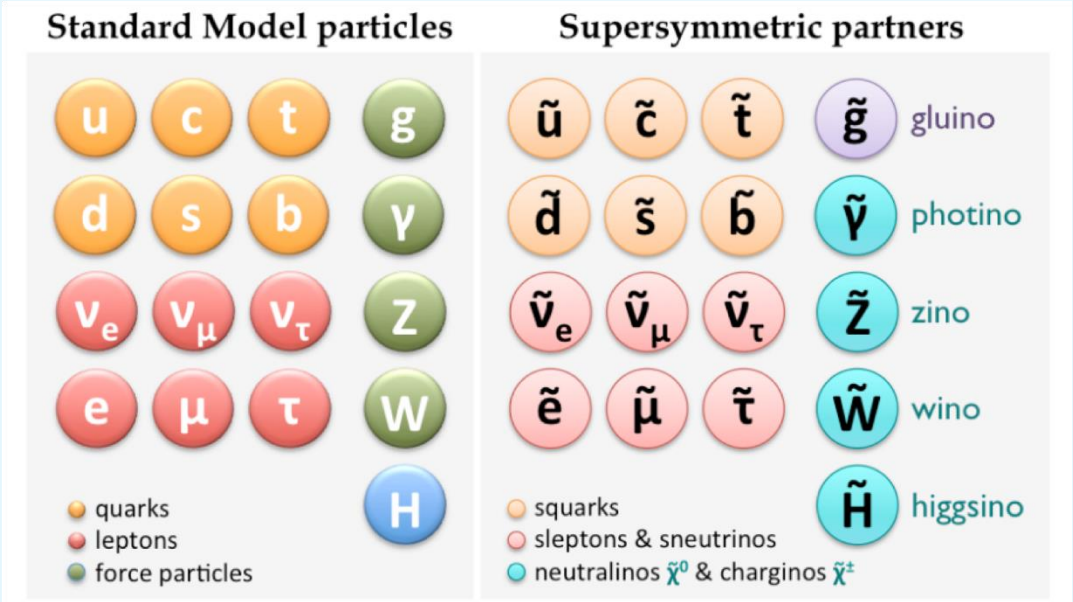
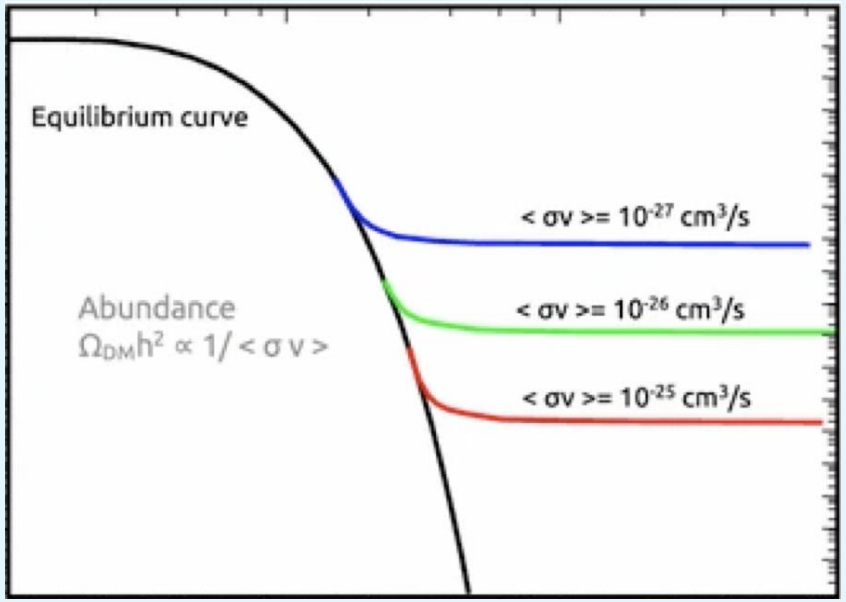
theoretical motivation

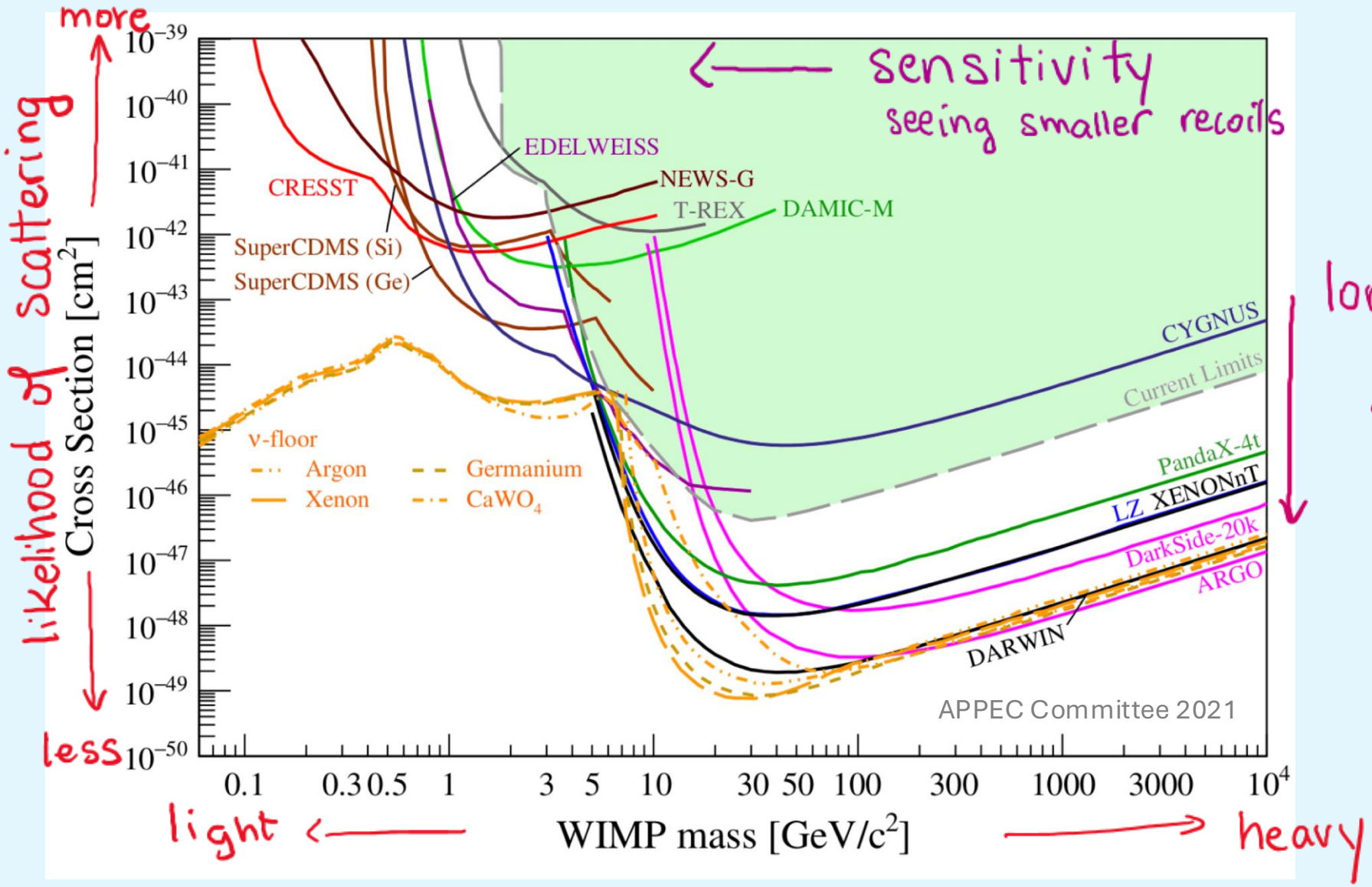


fits observations



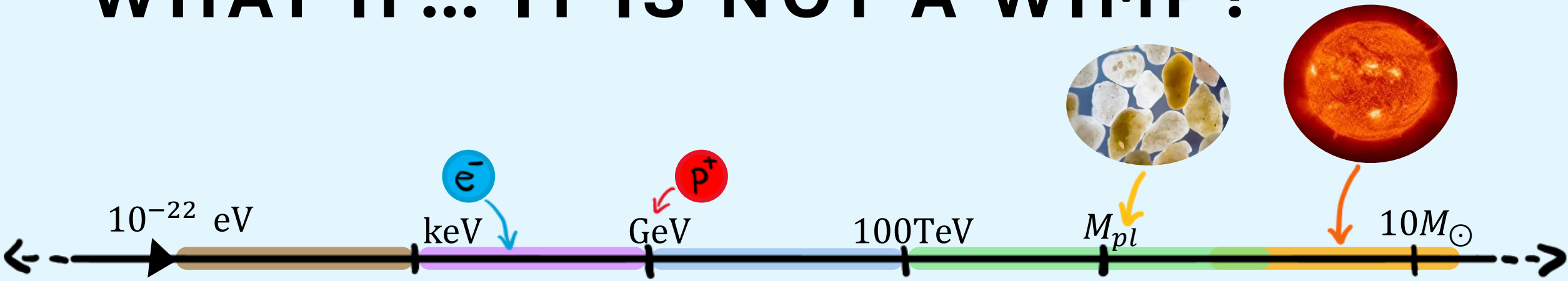
a miracle! hooray, a great reason to go look!

CURRENT STATUS OF WIMPS

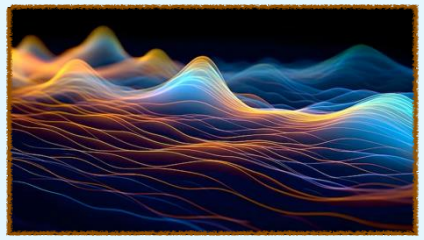


longer "exposure"
detector
size + runtime

WHAT IF… IT IS NOT A WIMP?



“ultralight” dark matter



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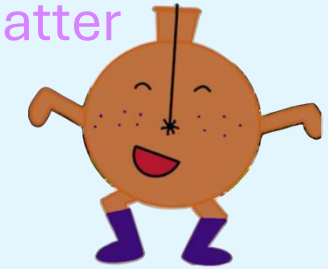
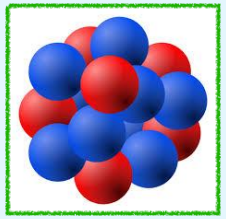
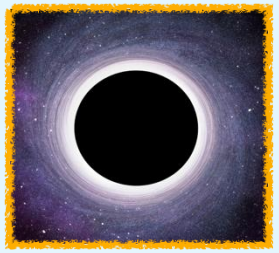


WIMPs

composite dark matter



primordial black holes



it is not the end!

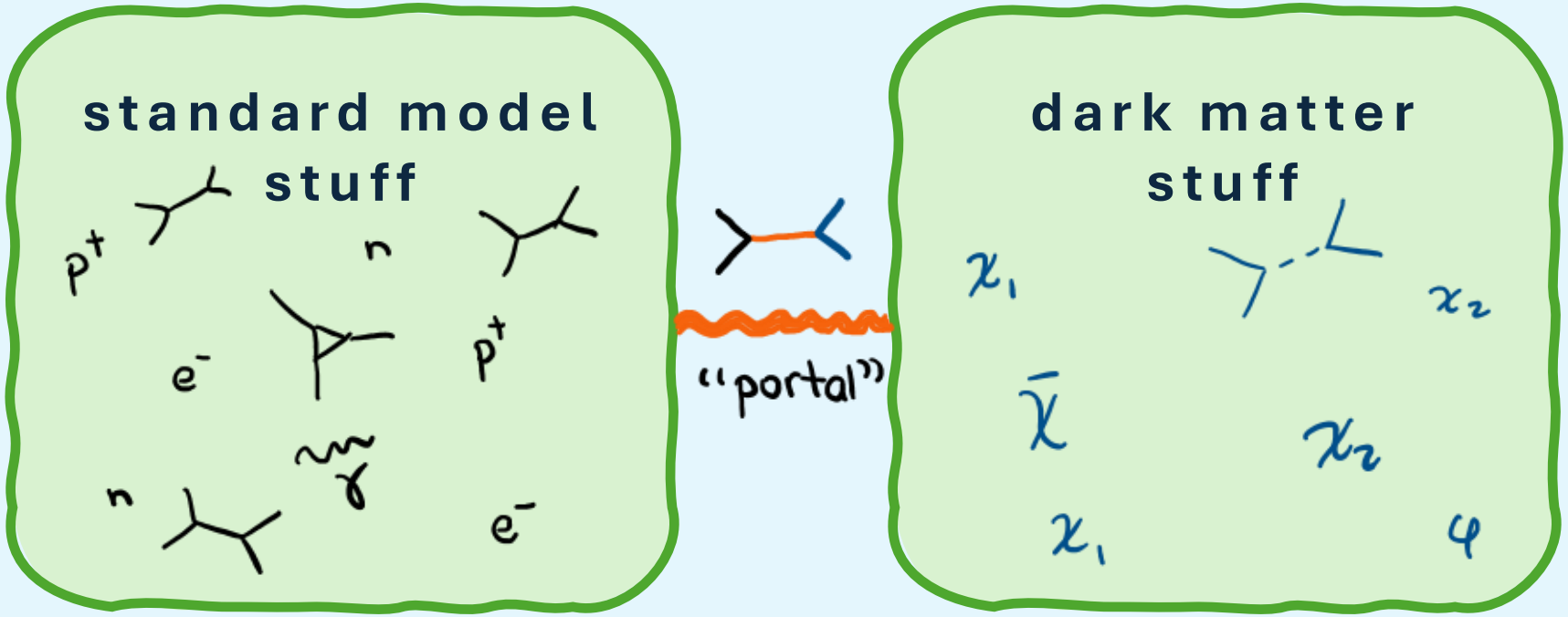
we just need to adapt our approach.

A HIDDEN DARK SECTOR

WEAK INTERACTION

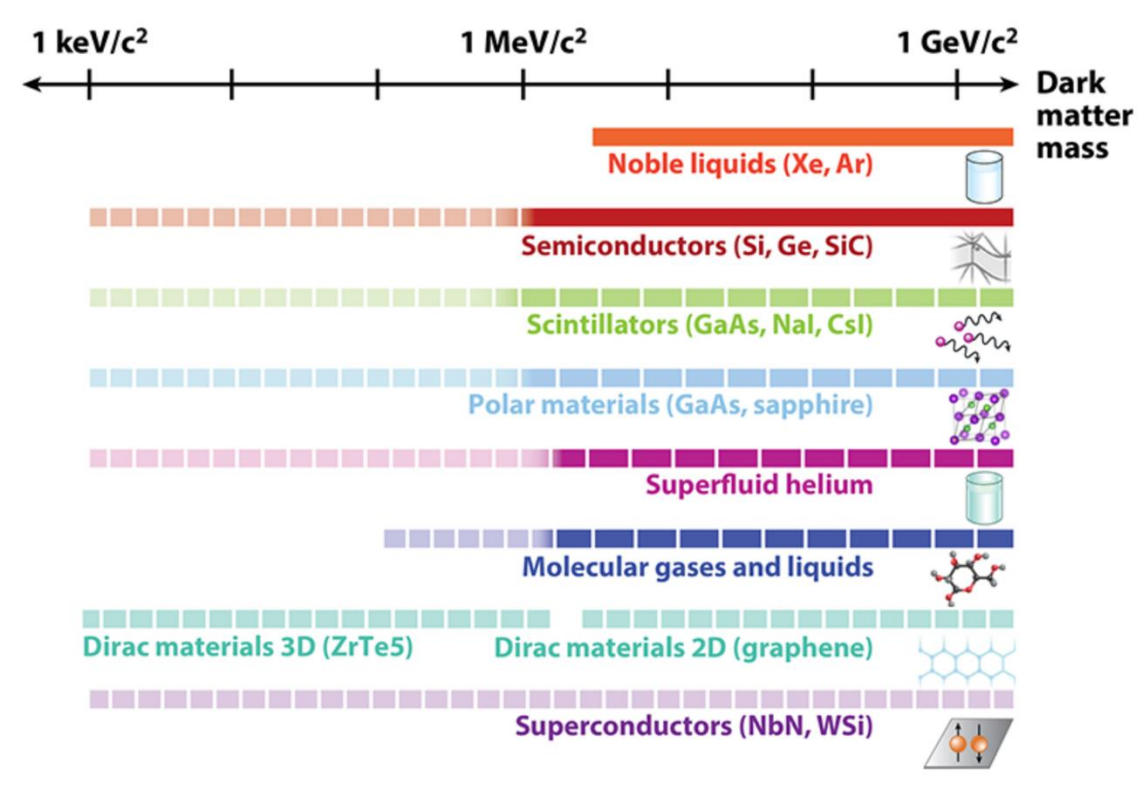
A HIDDEN INTERACTION

A HIDDEN DARK SECTOR

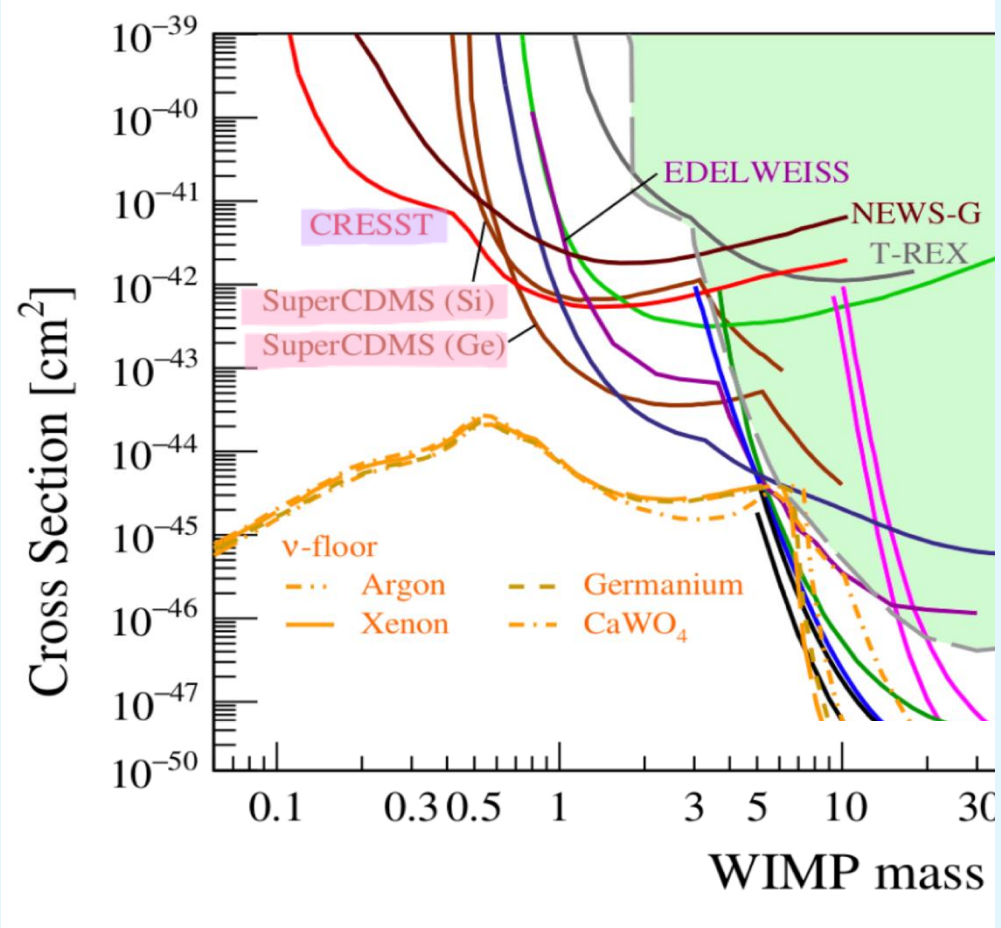


turns out, hidden sector DM could be lighter than the WIMP

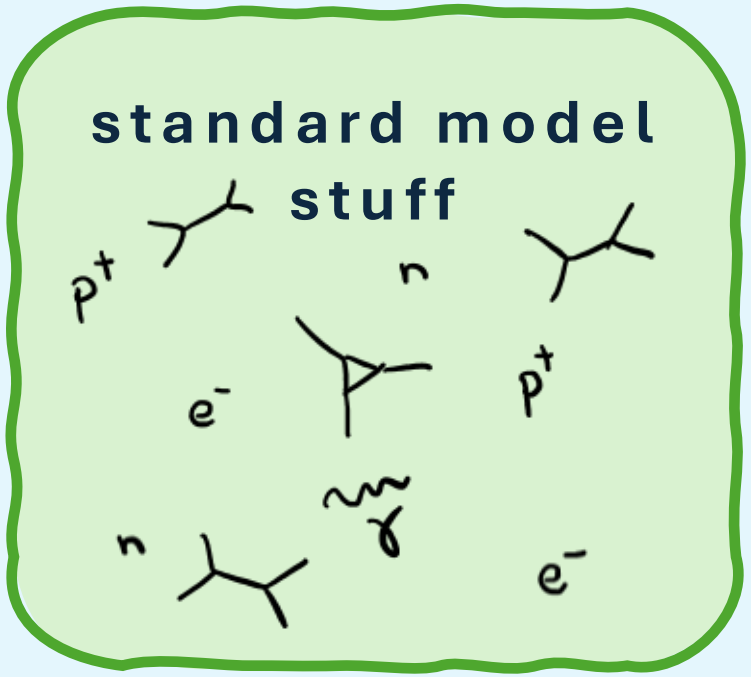
HIDDEN SECTORS, SUB-GEV DM

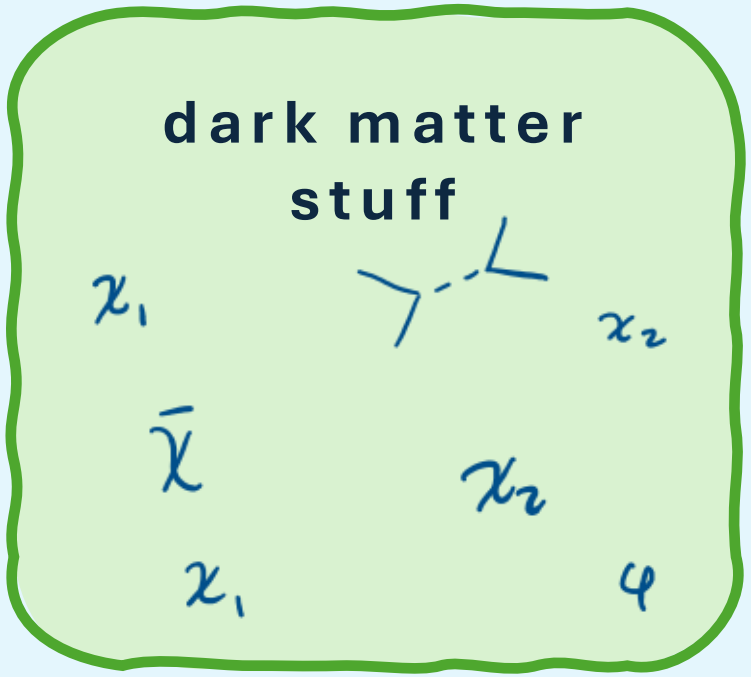


Essig 2020



WHAT IF THERE ARE NO INTERACTIONS?





there are reasons to suppose there is a "portal":
why are the amounts of DM and SM so close together?

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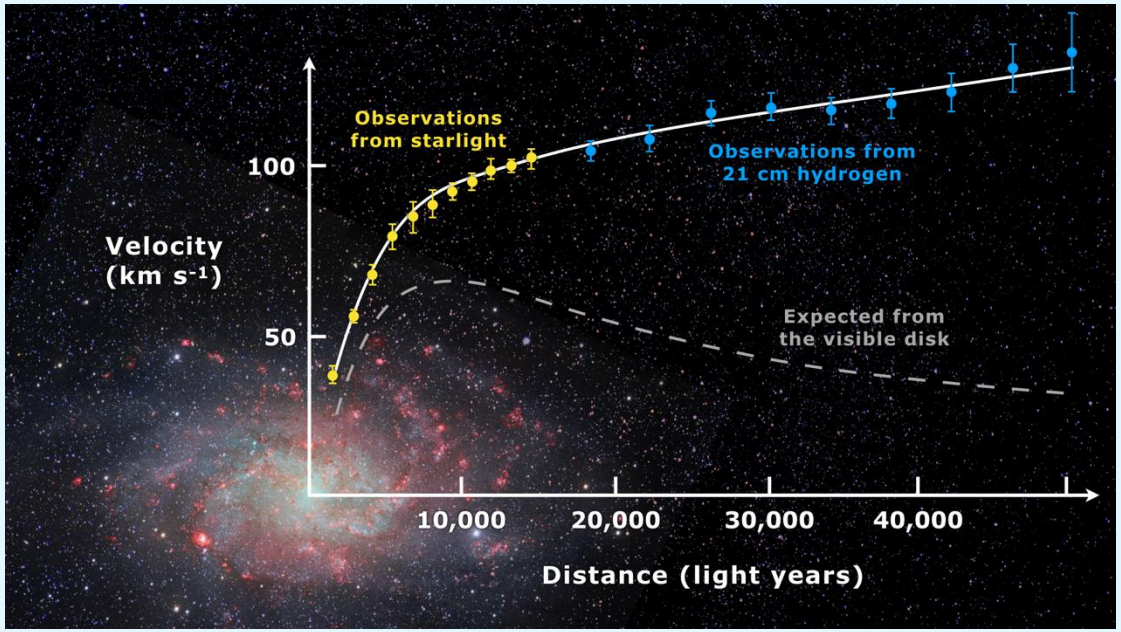
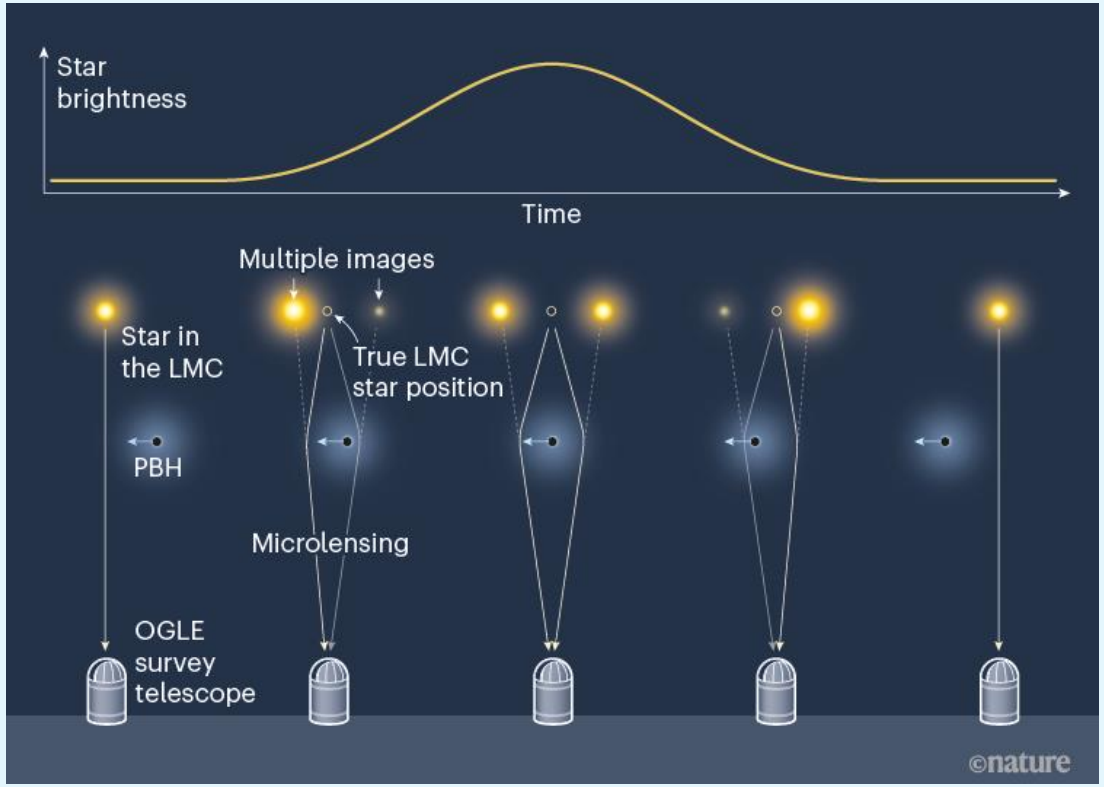
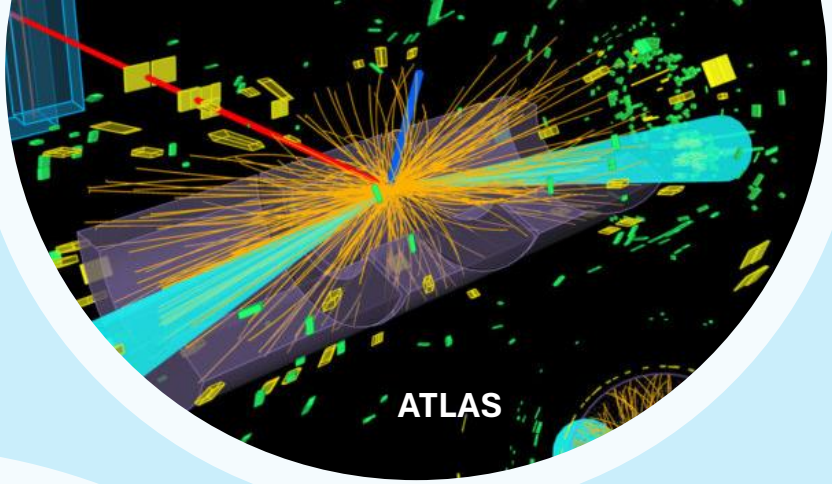
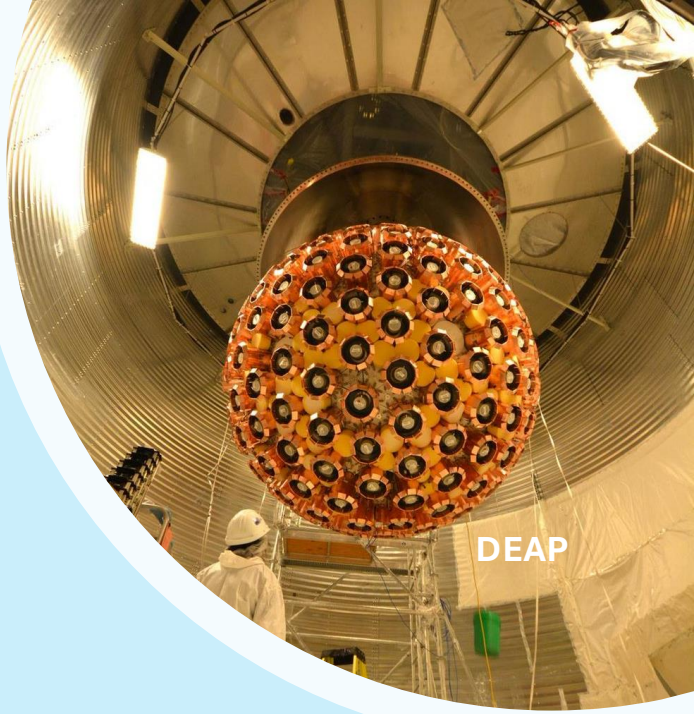


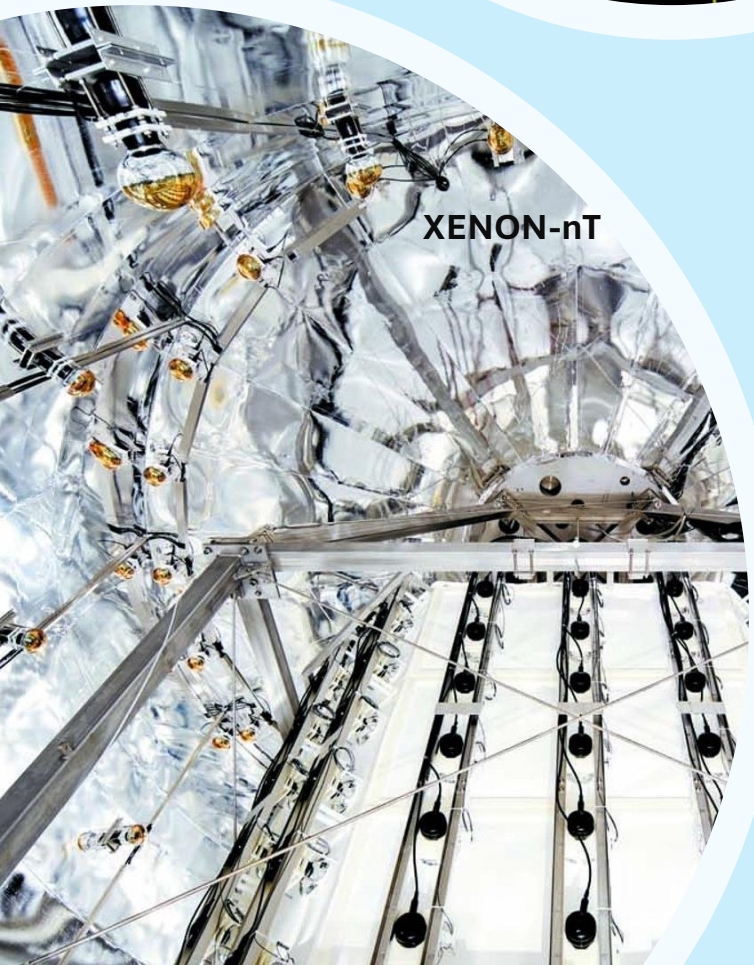
a first hint of dark matter came from gravity



We can search using gravitational probes







SO WE KEEP LOOKING!